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A comparison of Gaussian Diffusion and Flow Matching for Text to Motion generation

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Abstract — Human motion generation is a rapidly growing field, with a variety of applications. Generating realistic, text-accurate motion is an inherently difficult task, due to the many to many nature of the problem. Language can be ambiguous, motion can be described in many ways. In this paper, we present two well performing approaches to this problem and compare them based on many different metrics. The first approach is a Transformer-based Gaussian Diffusion model, that is adopted from Tevet et al. [2022]. The second model is developed by Hu et al. [2023]. In its core, it is a Conditional Motion Flow model. The uniqueness of the paper, is that both core architecture parts, so Motion Flow and Diffusion are inserted into the same shared backbone architecture. Both models use the same conditional setup, with the exception of the timestep embedding, which is discrete in diffusion and continuous in flow matching. The models are trained on the HumanML3D database from Guo et al. [2022]. After the comparison it will be clear that both approaches achieve comparable and good results in terms of FID and other metrics, while the ODE-based Motion Flow inference approach yields a significant, 7.7 times speedup.

1 Introduction

Text to Motion generation is a field that can have huge implications on many different fields. It offers the best "interface" for quick and potentially precise Motion synthesis. In Virtual Reality, Robotics, or Animations related environments, a precise and well behaving text-to-motion generation can mean a substantial speedup in the field, or even revolutionize it in the case of humanoid robotics.

This, however is a difficult task. This type of task is inherently many-to-many meaning, that one can describe a motion in many ways, and also my description can mean several things. The term kick can have several meanings depending on the context. Incorporating adjectives that are often necessary to annotate a movement greatly increase dimensionality. In addition movement has a temporal aspect, which makes quick inference difficult. Going further, one can understand what the movement is doing, but can tell that it is artificial. Problems such as foot-sliding, boundary violations and broken movement are all easy to spot for the human eye.

Throughout the years, many approaches were developed for well working Text to Motion synthesis. These generally coincided, with the current trend in Artificial Intelligence. In the early days of Motion generation, RNN based approaches such as Martinez et al. [2017] mainly focused on continuing an input motion and not on text conditioned generation. After the introduction of the Transformer architecture by Vaswani et al. [2017], Motion generation moved towards General Adversarial Networks (GAN) and Variational Autoencoders (VAE). Works, such as ACTOR by Petrovich et al. [2021] and Action2Motion by Guo et al. [2020] used action (discrete class label) conditioning and VAE architecture to achieve great results. Nowadays several strong approaches are used for Text to Motion generation. On one hand VAE are still present and offer a clean way to generate high fidelity motion. Works such as Petrovich et al. [2022], Guo et al. [2023] showcase the usefulness of VAE's in Motion generation.

Another common architecture choice is Diffusion, introduced by Ho et al. [2020]. Diffusion is an inherently good choice for many to many problems, such as image, video or motion generation, since throughout its many denoising steps, it can create very diverse outputs. Zhang et al. [2022] pioneered Diffusion in text to motion generation, while Tevet et al. [2022] introduced geometric losses to eliminate foot sliding and create physically more plausible motion.

A new approach in the Motion generation space is Motion flow. Hu et al. [2023] took inspiration from the foundational Flow Matching work of Lipman et al. [2023] and applied it to the motion domain. Rather than iteratively denoising over hundreds of steps, Flow Matching learns a continuous velocity field that transports noise to data via an ordinary differential equation (ODE). This means that at inference time, instead of

running through a thousand denoising steps, one can integrate the learned ODE in as few as ten steps. The practical implication is a dramatic reduction in inference time, while maintaining competitive motion quality. Whether this trade-off holds consistently against a well-tuned diffusion baseline, however, remains an open question. Comparing these two paradigms is not straightforward. Works in the literature evaluate their models on varying datasets, with different backbone architectures, different training durations, and different evaluation protocols. This makes it difficult to attribute performance differences to the generative engine itself, rather than to the many confounding factors surrounding it. A fair, controlled comparison, where the only variable is the generative engine, has not been conducted in the text-to-motion domain.

In this work, we address this gap directly. We embed both a Gaussian Diffusion engine and a Conditional Flow Matching engine into the same shared Transformer backbone, originally proposed by Tevet et al. [2022]. Both models are trained on the HumanML3D dataset by Guo et al. [2022], with identical conditioning, identical hyperparameters, and identical evaluation protocols. The only point of difference between the two models is the generative engine itself, and the treatment of the timestep, which is discrete in the case of Diffusion and continuous in the case of Flow Matching. This controlled setup allows us to isolate the effect of the generative engine and draw conclusions that are not muddied by external factors.

Our experiments reveal a meaningful trade-off between the two approaches. Flow Matching requires more training steps to reach peak performance, yet once deployed, it offers an approximately eight times reduction in inference time owing to the efficiency of ODE-based sampling. Both approaches achieve competitive results across standard evaluation metrics, with Flow Matching showing superior text-motion alignment as measured by R-Precision, and Gaussian Diffusion achieving a lower Fréchet Inception Distance. A notable limitation of Flow Matching, however, is a tendency to produce shaking or jittery motion, particularly in the local joints, which is perceptible to the human eye even when the global motion structure is correct. Neither approach dominates unconditionally, and the right choice depends on the demands and constraints of the application at hand.

2 Related Work

2.1 Early Sequence-Based Motion Generation

Human motion generation is a research field that has been around for quite a while, and its development is very much aligned with other deep learning trends. Initially, there was a strong focus on Recurrent Neural Networks. Martinez et al.'s Martinez et al. [2017] paper is a good representative of this time, where future frames from a given seed motion are predicted, rather than generating motion from scratch. These models performed well enough for a few time steps but started to average out and become blurry for longer sequences, a known problem with this type of approach. However, with the arrival of the Transformer Vaswani et al. [2017], this field was quick to adapt to this new technology, and its ability to attend over sequences of arbitrary length made it much easier to handle the kind of long-range dependencies that exist in human motion data, and most of the following research is a variation of this approach. ACTOR Petrovich et al. [2021] and Action2Motion Guo et al. [2020] are good representatives of this time, where a VAE is used in conjunction with a Transformer to condition human motion generation on a discrete action label such as walking or jumping. These models are still not text-conditioned, but they established the VAE as a good representation for human motion generation, a trend that continues in this text-conditioned version of this research field.

2.2 VAE-Based Text-to-Motion Models

The transition to natural language conditioning was a gradual one. TEMOS Petrovich et al. [2022] and T2M Guo et al. [2022] were two of the first to really tackle this problem well, learning a joint latent space between text and motion such that you could sample diverse motions given a text description. This was an important step forward. Since then, work in the VAE style has continued but in different ways. T2M-GPT Zhang et al. [2023a] was one of the first to use a VQ-VAE to quantize the latent space, generating motion tokens through an autoregressive GPT-style transformer. This was a strong paper, showing that thinking of motion as a sequence of language tokens was a good idea. AttT2M Cai et al. [2023] built on top of this, adding a body part aware quantization. This was an effort to capture more fine-grained structure in the motion. MoMask Guo et al. [2023] is a very strong paper in this line of work. They use a hierarchical

residual VQ-VAE with a masked transformer. This got an FID of 0.045 on HumanML3D, which is a very good result for this type of metric. MLD Chen et al. [2023] was another paper in this style, but from a slightly different angle. They used a VAE latent space but a diffusion model within this space, which significantly reduced inference times compared to a diffusion model in the raw space.

2.3 Diffusion-Based Motion Generation

Diffusion models, as popularized by the paper Ho et al. [2020], have become extremely popular in the motion generation space for good reason. The process of denoising does not make any assumptions about what the output should look like, and this makes it perfectly suited to the many-to-many nature of text-to-motion. MotionDiffuse Zhang et al. [2022] was the first to use this in the space of text-driven motion generation. They use a cross-modal transformer as their denoiser and show that fine control over specific parts of the human body and time intervals can be achieved in this space. MDM Tevet et al. [2022] built upon this work and is arguably the most important paper in this space. They use a clean transformer denoiser with classifier-free guidance Ho and Salimans [2022]. They also add geometric losses for foot contact and joint velocity, and this makes the generated motions much more realistic. This paper is the direct continuation of MDM, as the modularity of this work is what allows us to replace the generative engine while leaving almost everything else the same. After MDM, much of the work in this space has been trying to solve the problem of inference speed. ReMoDiffuse Zhang et al. [2023b] attempted to make the generated motion-text pairs more consistent by conditioning the denoiser on similar motions retrieved from a database. MMM Pinyoanuntapong et al. [2024] also attempted to solve this problem by using discrete motion tokens and a masked diffusion transformer. MotionLCM Dai et al. [2024] also attempted to solve this problem by using consistency distillation to get their generation speed closer to real time. All of these papers are interesting in their own right but are essentially workarounds to the problem.

2.4 Flow Matching for Motion Generation

Flow Matching Lipman et al. [2023] is a relatively newer generative approach that takes a completely different route. Instead of learning the reverse of a noise process, it learns a continuous vector field that maps from the noise distribution to the data distribution along an ODE trajectory. The advantage here is that the trajectory is much straighter than the stochastic trajectory that the diffusion models take and hence can be integrated in many fewer steps, just ten steps rather than the hundreds that the diffusion models need to take. The training objective is also much simpler, just a regression to a target velocity field without a complex variance schedule. The paper by Hu et al. Hu et al. [2023] introduced flow matching into the motion generation space and achieved competitive results with HumanML3D and KIT-ML with much faster inference times. They also introduced a motion editing technique called sampling trajectory rewriting that utilizes the deterministic trajectory for motion in-betweening and upper body editing, which is a nice capability that the diffusion models do not have access to.

2.5 Motivation and Research Gap

Yet with all this progress, there is a gap in the literature that this paper hopes to fill. The comparisons that have been done between the diffusion and the flow matching that have been done so far are not apples-to-apples due to the different architectures and training and test data that have been used for the experiments. It is not clear with confidence which engine is actually better for the task at hand. We address this directly in this paper by taking the backbone of the MDM model that was introduced in Tevet et al. [2022] and replacing the diffusion engine with a Conditional Flow Matching engine that was introduced in Hu et al. [2023] and keeping the remainder of the models identical and training both models on the HumanML3D dataset introduced in Guo et al. [2022] and evaluating the models with the standard metrics of R Precision, FID Score, and Diversity Score.

3 Methods

The idea behind the work is very simple: start with the backbone from the MDM paper Tevet et al. [2022] and replace the diffusion engine with a Conditional Flow Matching engine, also known as CFM, from the Hu et al. [2023] paper, leaving all the rest of the model completely intact. This section explains the common architecture, and then the differences between the two generative engines and their training and sampling procedures.

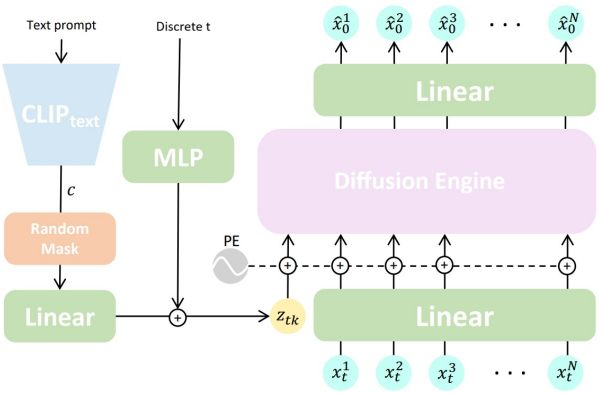

Figure 1 Figure shows the underlying architecture of the Diffusion setting. The figure is from Tevet et al. [2022] with the changed t input and Transformer name.

3.1 Shared Backbone Architecture

Both models are based on the same Transformer backbone, an encoder-only variant. The architecture follows the MDM paper directly. The input to the model during a forward pass consists of a noisy motion sequence x_t, a timestep t, and a conditioning signal c. The conditioning signal is a text embedding from a frozen CLIP ViT-B/32 encoder Radford et al. [2021]. The timestep t and the condition code c are both projected to the Transformer dimension via separate feed-forward networks and summed to give a single input token z_t^k. Each frame in the motion sequence is linearly projected to the Transformer dimension and summed with a standard sinusoidal positional embedding. The token z_t^k and the motion frame tokens are input to the Transformer encoder. The first output token corresponding to z_t^k is discarded, and the rest are projected back to the original motion dimension to give the output. The Transformer backbone consists of 8 layers and a latent dimension of 512. Both models are trained with classifier-free guidance Ho and Salimans [2022] by stochastically dropping the conditioning signal for 10% of training samples. This forces the model to learn both the conditioned and unconditional distributions from the same set of weights. At inference, the guidance scale is set to $s = 2.5$ for both models, which is a good balance between diversity and fidelity. The motion representation is based on HumanML3D Guo et al. [2022] and consists of a concatenation of root velocity, joint positions, joint velocities, joint rotations, and binary foot contact labels. This is shared between both models.

3.2 Gaussian Diffusion

The Gaussian Diffusion engine is based on the DDPM formulation of Ho et al. [2020]. A clean motion sample x_0 is progressively noised over $T = 1000$ discrete timesteps with a fixed cosine noise schedule, resulting in a series of noisier samples $x_1, x_2, \ldots, x_T$. The forward process is defined by

$$q(x_t|x_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}\, x_{t-1},\ (1-\alpha_t)I), \qquad (1)$$

where $\alpha_t \in (0,1)$ are fixed hyperparameters. Instead of predicting noise ϵ as in the original DDPM, MDM directly predicts the clean sample $\hat{x}_0 = G(x_t, t, c)$, following Ramesh et al. [2022]. This is what allows geometric losses to be tractable in this framework. The training objective is

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{x_0,\, t}\, \|x_0 - G(x_t, t, c)\|_2^2\,. \qquad (2)$$

On top of this, three geometric losses are added: a position loss $\mathcal{L}_{\text{pos}}$ to penalize errors in forward-kinematic joint positions, a foot contact loss $\mathcal{L}_{\text{foot}}$ to nullify foot velocities in case of ground contact, and a velocity loss $\mathcal{L}_{\text{vel}}$ that penalizes errors in frame-to-frame joint velocities. The total training loss is

$$\mathcal{L} = \mathcal{L}_{\text{simple}} + \lambda_{\text{pos}}\mathcal{L}_{\text{pos}} + \lambda_{\text{vel}}\mathcal{L}_{\text{vel}} + \lambda_{\text{foot}}\mathcal{L}_{\text{foot}}. \qquad (3)$$

The timestep t is discrete, uniformly sampled from $\{1, \ldots, T\}$ during training, and embedded via a sinusoidal embedding before projection. The sampling process runs iteratively over all T steps, predicting $\hat{x}_0$ at each step and diffusing it back to x_{t-1}, running from $t = T$ down to $t = 1$.

3.3 Conditional Flow Matching

The Flow Matching engine substitutes the discrete denoising process with a continuous one. Following Lipman et al. [2023] and Hu et al. [2023], the model learns a velocity field $v(x_t, t, c;\, \theta)$ that defines a flow between a Gaussian noise distribution $p_0 = \mathcal{N}(0, I)$ and the data distribution $p_1 \approx q(x)$ via a neural ODE

$$\dot{\phi}_t(x) = v(\phi_t(x),\, t), \quad \phi_0(x) = x_0. \qquad (4)$$

The conditional probability path is defined as a linear interpolation

$$x_t = t\, x_1 + [1 - (1 - \sigma_{\min})\, t]\, x_0, \qquad (5)$$

where $\sigma_{\min} = 10^{-4}$ is a small residual noise term that keeps the path well-conditioned near $t = 1$. The corresponding target vector field is

$$w_t(x \mid x_1) = x_1 - (1 - \sigma_{\min})\, x_0, \qquad (6)$$

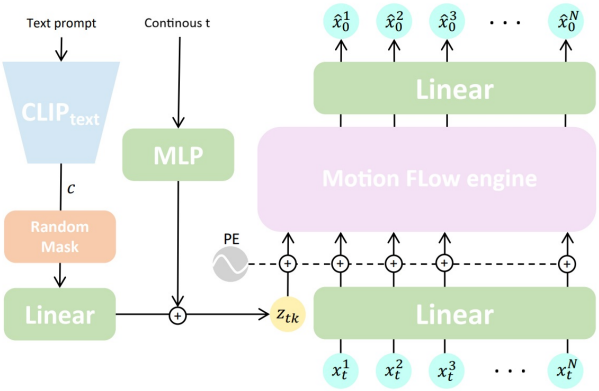

Figure 2 Figure shows the underlying architecture of the Motion Flow setting. The figure is from Tevet et al. [2022] with the changed continuous t input and Transformer name.

and the training objective is a simple regression

$$\min_{\theta} \mathbb{E}_{t,\, x_t,\, x_1} \|v(x_t, t, c;\, \theta) - w_t(x \mid x_1)\|_2^2 \,. \qquad (7)$$

The timestep t is continuous, uniformly sampled from $[0, 1]$ during training, and embedded via the same sinusoidal embedder as in the diffusion model, adapted to accept continuous values. At inference, the learned velocity field is integrated using an Euler ODE solver discretized over $N = 10$ steps, reducing the number of network evaluations from 1000 in the diffusion model to just 10. The geometric losses from the diffusion model are retained in the flow matching training objective, evaluated on the predicted $\hat{x}_1$ in the same way.

The only differences between the two models are therefore the generative engine, the training objective, and the treatment of the timestep. Everything else — the Transformer backbone, the CLIP text encoder, the classifier-free guidance scheme, the motion representation, the dataset, the batch sizes, and the evaluation metrics — is kept fixed. This ensures that any observed difference in performance can be attributed cleanly to the choice of generative engine, and not to any confounding factor in the surrounding architecture or training setup.

4 Experiments

Both models were trained using a single RTX 5070 (12 GB) GPU on the HumanML3D dataset Guo et al. [2022], which comprises 14,616 motion sequences and 44,970 text annotations. Following preprocessing and mirroring-based augmentation, the final working set comprises 26,846 sequences and 87,387 annotations. The Gaussian Diffusion model and the Flow Matching model were trained with a batch size of 64. Both models share the same conditioning, evaluation protocol, and geometric loss weights. The evaluation protocol employs the standard HumanML3D test set and reports R-Precision (top-1,top-2 and top-3), FID and Multimodality and Diversity. All results are computed across multiple independent replications and reported with 95% confidence intervals derived from the t-distribution.

4.1 Diffusion engine

The diffusion model converges slowly to its best FID, which is obtained at around 400K steps. At 100K steps the model achieves a top-3 R-Precision of 0.594±0.008 and a FID of 1.495 ± 0.275, already competitive for text-motion alignment but still far from its generative peak. By 200K steps the R-Precision has dropped to 0.557 ± 0.006 (FID 0.545 ± 0.044), a counter-intuitive result that suggests the model is going through a period of re-organization in its internal representations before settling into more stable behavior. The best FID observed, 0.407 ± 0.049 across 20 replications, occurs at 400K steps. At 500K and 540K steps the FID climbs back to 0.552 and 0.681 respectively, indicating that the diffusion model does not benefit from running beyond its peak and that training must be stopped at the right moment according to this metric.

The R-Precision of the diffusion model stabilizes between 0.555 and 0.566 for all steps after 200K, never returning to the 0.594 seen at 100K. The ceiling for top-3 R-Precision appears to be around 0.57 regardless of training length, pointing to a fundamental limit in how well the DDPM objective translates conditioning signals into the generated output.

4.2 Motion Flow engine

The Flow Matching results tell a very different story. By 100K steps top-3 R-Precision reaches 0.625 ± 0.009, and by 200K steps it climbs to 0.644 ± 0.030, where it plateaus. From 200K to 540K steps, top-3 R-Precision moves only from 0.644 to 0.632, a change well within noise. Text-motion alignment saturates early and additional training does not help it.

FID, by contrast, improves steadily across the entire training run. At 300K steps, with 20 replications, FID reaches 0.380 ± 0.037. At 540K steps, again with 20 replications, it reaches 0.353 ± 0.033 — the lowest FID recorded across all flow matching checkpoints. Unlike the diffusion model, Flow Matching does not overshoot and regress; the FID curve is monotonically decreasing across every checkpoint evaluated.

Table 1 Top-3 R-Precision and FID across training checkpoints. Confidence intervals are 95% via t-distribution. † marks the checkpoint of best FID for each model. **Bold** indicates the best value across all models and checkpoints.

Model & steps	top-3 R-Prec	FID
Diffusion 100K	0.594 ± 0.008	1.495 ± 0.275
Diffusion 200K	0.557 ± 0.006	0.545 ± 0.044
Diffusion 300K	0.532 ± 0.007	0.731 ± 0.053
Diffusion 400K†	0.564 ± 0.008	0.407 ± 0.049
Diffusion 500K	0.555 ± 0.009	0.552 ± 0.044
Diffusion 540K	0.548 ± 0.007	0.681 ± 0.068
Flow Match. 100K	0.625 ± 0.009	0.475 ± 0.042
Flow Match. 200K	**0.644 ± 0.030**	0.399 ± 0.103
Flow Match. 300K	0.638 ± 0.007	0.380 ± 0.037
Flow Match. 400K	0.635 ± 0.006	0.374 ± 0.031
Flow Match. 500K	0.632 ± 0.007	0.375 ± 0.030
Flow Match. 540K†	0.632 ± 0.007	**0.353 ± 0.033**

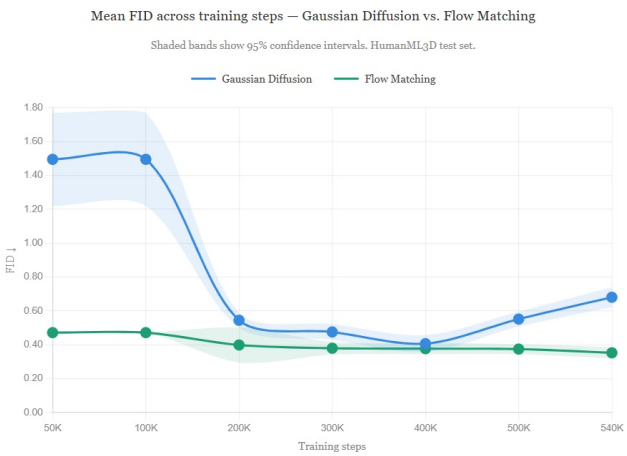

Figure 3 The figure shows the FID progression for the two trained models at different training steps. The shaded bands around the progression line represent the 95% confidence intervals.

4.3 Head-to-head comparison

Table 1 reports all evaluated checkpoints. The cleanest comparison is at 200K steps, where both models have seen similar amounts of data. Flow Matching achieves top-3 R-Precision 0.644 ± 0.030 against diffusion's 0.557 ± 0.006, a gap of 15.6 percentage points, and FID 0.399 ± 0.103 against diffusion's 0.545 ± 0.044. At this matched point Flow Matching leads on both metrics simultaneously. The situation changes as diffusion trains longer. Diffusion reaches its best FID of 0.407 ± 0.049 at 400K steps; at that same point Flow Matching at 300K already records FID 0.380 ± 0.037. The divergence is most striking at 540K steps, where both models have been trained for the same number

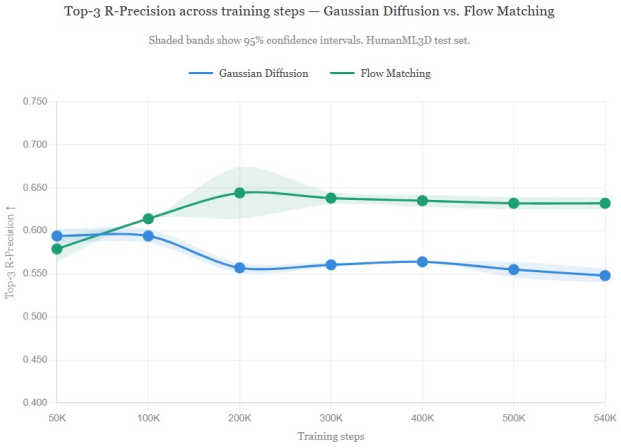

Figure 4 The figure shows the top 3 R-score progression for the two trained models at different training steps. The shaded bands around the progression line represent the 95% confidence intervals.

of steps and each has 20 replications. Flow Matching records FID 0.353 ± 0.033 and top-3 R-Precision 0.632 ± 0.007. The diffusion model at the same checkpoint records FID 0.681 ± 0.058 and top-3 R-Precision 0.548 ± 0.008. The diffusion model has deteriorated well past its 400K peak, while Flow Matching is still at its best. This asymmetry is not a marginal difference: the FID gap at 540K is 0.328 in favour of Flow Matching, compared to a gap of just 0.033 at 400K versus 300K. Continued training actively harms the diffusion model while leaving the flow model unaffected.

Inference time is unambiguous. The velocity field, integrated with an Euler solver over 10 ODE steps, takes approximately 6 seconds per batch on the RTX 5070. The diffusion model's DDPM sampling over 1000 steps requires approximately 43 seconds per batch — a 7.7× wall-clock speedup in favour of Flow Matching.

4.4 Qualitative observations

While these quantitative metrics may not capture all aspects of model performance, it is possible for a model to have high FID scores and yet not produce motion that is perceptually realistic. Similarly, it is possible for a model to have low R-Precision and yet produce motion that is perceptually realistic. We therefore supplement this quantitative information with observations about the visual quality of the motion generated.

As a reference case, it is clear that the original MDM generations in Tevet et al. [2022] are conservative in their motion. The model does not generate motion unless it is needed according to the text prompt. In particular, limbs that do not need to move according to

Table 2 Quantitative results on HumanML3D. Values for MDM and MFM are taken from Tevet et al. [2022] and Hu et al. [2023] respectively. Our models are evaluated across multiple independent replications; confidence intervals are 95% via t-distribution. ↓ lower is better; ↑ higher is better; → closer to real is better.

Method	NFE ↓	RP top-3 ↑	FID ↓	MM-Dist ↓	Diversity →
MDM Tevet et al. [2022]	1000	$0.611^{\pm.007}$	$0.544^{\pm.044}$	$5.566^{\pm.027}$	$9.559^{\pm.086}$
MFM Hu et al. [2023]	10	$0.642^{\pm.003}$	$0.362^{\pm.006}$	$5.280^{\pm.009}$	$9.860^{\pm.095}$
Ours: Diffusion 400K	1000	$0.564^{\pm.008}$	$0.407^{\pm.049}$	$4.470^{\pm.044}$	$7.622^{\pm.089}$
Ours: Diffusion 540K	1000	$0.548^{\pm.008}$	$0.681^{\pm.058}$	$4.622^{\pm.029}$	$7.503^{\pm.077}$
Ours: Flow Match. 540K	10	$0.632^{\pm.006}$	$0.353^{\pm.030}$	$4.124^{\pm.030}$	$8.265^{\pm.073}$

the text prompt do not move. The motion is generally stable and perceptually realistic. The only recurring error is left-right confusion, where the model gets the direction of motion wrong but produces semantically correct motion nonetheless.

Our own diffusion model follows a path that is only partially reflected in the quantitative results. At 100K steps, simple motion such as walking forward, sitting down, and raising an arm is already rendered convincingly. More complex motion is still not well handled at this stage, but even in those cases where motion is successfully generated, it is clean and physically realistic. As training progresses, more and more motion is successfully captured. At 540K steps, the perceptual quality of the generations is substantially better than it was at 340K, even though the FID metric is worse than it was at 400K. This divergence is a real effect. The metric appears to be measuring something that diverges from perceived realism in the late stages of training, which tells us that FID alone is not sufficient as a stopping criterion. Interestingly Stein et al. [2023] and Jayasumana et al. [2024] also note this phenomenon about generative model evaluation.

The flow matching model presents the opposite problem at early checkpoints. Although R-Precision and FID scores already favor Flow Matching over Diffusion at 100K steps, the sequences produced at this point are very difficult to interpret. This is due to very strong per-joint jitter: the global motion structure is correct, the body moves in the right direction and at a reasonable velocity, but individual joints oscillate rapidly from frame to frame in a manner that obscures the intended motion and appears completely unnatural. This is consistent with ODE discretisation error accumulating across joints when the velocity field is still early in training Zhou and Liu [2025]. The discrepancy with the metrics at early checkpoints is therefore somewhat misleading. If shown the 100K flow matching and diffusion sequences side by side, a viewer would likely prefer the diffusion output, despite the metric results suggesting otherwise.

The jitter diminishes as training progresses. By 540K steps it is no longer the dominant perceptual artifact, and the generated motions are clear and, for the most part, believable. A further observation from the flow matching evaluation is that the choice of ODE solver and step count has no discernible effect on the output. Sequences generated with the Euler method at 10 steps are virtually indistinguishable from those generated with a higher-order method and more steps. This indicates that the learned velocity field is smooth enough that coarse integration is sufficient Lipman et al. [2023], Hu et al. [2023].

4.5 Comparison with baselines

The most interesting findings are that the newly trained Motion Flow achieved similar FID 0.353 ± 0.033, which is lower than the original MDM and similar to the original MFM 0.362 ± 0.006 by Hu et al. [2023]. This is significant, because the model was put into an architecture that although was suited for it, was not designed in mind with the use of a Conditional Flow Matching engine as the main module. This shows, the power of the embedding, showing that the conditional z_t^k is flexible enough to serve both architectures. In addition the R-score tells a similar story, reaching a score of 0.632 ± 0.006 top-3 that is well into the range of Tevet et al. [2022] and Hu et al. [2023].

The MM-Dist figures measures the distance between generated motion and text embeddings, so a lower value indicates better text-motion alignment in embedding space. For all cases our models achieved a worse score than either original paper.

Taken all together, the results indicate, that putting the Conditional Flow engine into the MDM architecture yields a promising model. It achieves good FID and R-score values, comparable to the original paper and beating that of the retrained Diffusion Model at

540K steps, while requiring significantly less steps at inference (10 vs. 1000).

5 Conclusion

This paper aimed to provide an answer to a deceptively simple question: if we replace the generative model in text-to-motion with Conditional Flow Matching instead of Gaussian Diffusion, with all other components remaining the same, then what changes? The answer, in the end, was more obvious than the mixed signals in the prior work might have led us to expect.

Flow Matching is the better performing model in virtually all quantitative ways. It trains faster, with competitive FID and R-Precision scores far earlier in the process. It also has better text-motion alignment during training, with a top-3 R-Precision gap of 8-10 percentage points at the stages after 100K steps. Its FID score improves monotonically rather than peaking and falling like the diffusion model. And at test time, it only needs to take 10 steps with its ODE instead of the diffusion model's 1000 steps, a 7.7× speedup in wall-clock time that is not a distillation trick but a structural result of the generative model framework. At 540K steps with 20 replications each, the FID score gap is 0.353 vs. 0.681, a result where the gap between models is large enough to be unambiguous.

The only area where things get complicated is in the qualitative stage. While Flow Matching surpasses the diffusion model in all quantitative metrics from 100K steps onward, in the same period its motion output appears visually jittery to the human eye, and less convincing than the diffusion model's output. While the metrics disagree with our visual impression, our findings are still relevant to the problem at hand, because FID and R-Precision are not reliable proxies for perceptual quality in the early stages of flow matching training Jayasumana et al. [2024], Stein et al. [2023]. By 540K steps, the jitter has dissipated to some extent, but remains an easily spotted issue.

One final point of validation comes from the original MDM and MFM papers. Our flow matching model, running in a framework not originally designed for it, achieves at least the same or better results than the original papers on FID and R-Precision at 10 NFE. This lends further weight to the efficacy of the MDM backbone as a platform for substitution of the underlying engine, and suggests that the conditioning mechanism — the shared z_t^k token — is flexible enough to work with both discrete and continuous time parameterizations.

What has not been accomplished by this work also bears mentioning. The Motion Flow model was trained to full convergence by the standards set by the original MFM paper Hu et al. [2023]. The Multimodality metric was not evaluated, and longer training runs would strengthen the conclusions. The phenomenon of jitter in the early stages of flow matching also deserves further investigation, as it may be possible to suppress it with better ODE solvers, modified training regimens, or adjusted geometric losses.

While Flow Matching may not be a drop-in replacement for diffusion in every setting, the results presented here make a strong case that it is very competitive and if the jitter issue could be solved by further training, or alternative architecture, then Flow Matching could achieve an unambiguous edge.

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